# Seamus Taylor

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## PROFILE

Third-year Computer Science student at Monash specialising in Data Science, with two years building applied AI systems and standing up the teams that deliver them. Shipped an LLM extraction pipeline with its own evaluation harness; architecting a clinical decision agent that retrieves from published guidelines and grounds every output in traceable evidence; leading the vision pipeline for an autonomously controlled prosthetic arm. Team Lead of Monash AIM for 2027. Most of my work has been founding something rather than joining it — setting architecture, scope and standards for teams that had none yet.

## EDUCATION

**Bachelor of Advanced Computer Science (Honours)** — Monash University *2024 – 2027 (expected)*
Specialising in Data Science; minoring in Mathematics.

**AI consulting internship, Claude Certified Architect** — AI consultancy, Melbourne *Nov – Dec 2026*
Six-week consulting placement; the certification covers industry-standard practice for RAG and production AI systems.

## EXPERIENCE

**Team Lead (2027), Operations Lead (2026)** — Monash AIM — Analysis of Images in Medicine *2025 – Current*
Monash's medical AI research group. We enter MICCAI, the leading international medical imaging conference, whose challenges are scored on held-out test sets against research labs worldwide.

***CHIMERA-Agent — clinical decision agent***

- Building a Python risk-stratification pipeline over MRI and biopsy data, architected as a retrieval-grounded hybrid: a deterministic guideline decision core handles the clinical logic, and the LLM is constrained to what it can actually support from the retrieved record.
- Risk-group derivation retrieves against published clinical guidelines and returns candidate categories with provenance traces rather than hard labels, so every output can be audited back to the source that produced it — the difference between a system you can deploy and one you can only demo.

***CohortX — LLM extraction pipeline (Team Lead, four people, three tasks)***

- Led delivery of a pipeline extracting structured cohort selection criteria from biomedical free-text, and personally delivered two of the three tasks end to end: parsing, prompting, eligibility prediction, evaluation and submission.
- Selected open-weight models — LLaMA 3.1 8B as primary, Mistral Nemo and BioBERT as fallbacks — evaluated against accuracy, inference cost and the compute actually available.
- Built a local evaluation harness reimplementing the challenge's composite scoring metric, so the team could measure a change without burning a submission, and diagnosed two silent faults: a parsed-versus-expected file mismatch, and scorer weights that could not be verified.
- Restructured ownership and handover so the team kept delivering to deadline through a period of absence. The third-task result earned a contribution to a peer-reviewed MICCAI paper and presentation, in preparation.

***Programme operations***

- Coordinated five parallel challenge teams — allocation, delegation, deadline tracking, liaison with academic advisors — and fitted training workloads to a zero-budget constraint across Monash M3/MASSIVE HPC and free Kaggle and Colab tiers. One of the five won its challenge outright.

**AI Lead, Computer Vision — NeuroGrip** — Monash DeepNeuron *2026 – Current*

- Own the computer vision pipeline on a three-person build (AI, mechanical, electrical) working toward a prosthetic arm controlled autonomously by AI; my module detects the target object and drives finger movement from what the camera sees. Lead the AI team from next semester as the project scales, owning architecture and scope.
- Built monocular depth estimation on Depth Anything with an optional stereo path and the calibration for both; the pipeline tracks a taught fingertip and the target, estimates the gap in millimetres, and emits a movement direction to the electrical controller over a shared-memory and Ethernet bus.
- Deployed to an NVIDIA Jetson NEON smart camera for on-device inference; wrote synthetic scene generation so the pipeline could be tested and regression-checked before any hardware existed.
- Enforced a safety interlock as a design rule: no actuator command is sent until calibration and validation pass, so a miscalibrated model cannot drive a physical hand.

**Research Project — EEG-to-fMRI prediction (NeuroBOLT)** — Monash, FIT2082 *2026*

- Assessed the feasibility of transformer-based EEG-to-fMRI prediction on a large paired neuroimaging dataset, reimplementing the NeuroBOLT architecture (NeurIPS 2024) from scratch under the supervision of a PhD researcher and an AI academic.

## SKILLS

**Languages:** Python, C++, SQL, R, basic Java
**AI engineering:** Open-weight LLMs (LLaMA, Mistral, BioBERT), retrieval-grounded generation with provenance, structured output, hybrid deterministic/LLM design, evaluation harnesses
**ML & systems:** PyTorch, transformers, CNNs, depth estimation, edge inference (Jetson), data pipelines, M3/MASSIVE HPC, Linux, Git
**Leading:** Team and operations leadership across five concurrent projects; Web Developer, Centaur Association, since Feb 2025